

23 Bias of Estimators

- MLE is one procedure for finding an estimator of a parameter θ . But when several estimators of θ are available, how do we decide which is “better”?
- The value of a parameter θ is unknown, so it is impossible to determine if any single estimate of θ is good or bad.
- An estimator is a random variable that returns different estimates of θ for different samples. So even if an estimator produces “good” estimates of θ for some samples, it might produce “bad” estimates of θ for others.
- Therefore, we can never determine for any particular sample if an estimator produces a good estimate of the true unknown value of θ .
- Rather, we evaluate the estimation *procedure*: Over many samples, does an estimator tend to produce reasonable estimates of θ for a variety of potential values of θ ?
- Remember: a statistic/estimator is a random variable whose sampling distribution describes how values of the estimator vary from sample-to-sample over many (hypothetical) samples.
 - We can estimate the degree of this variability by *simulating many hypothetical samples* from an assumed population distribution and computing the value of the statistic for each sample.
 - However, *in practice usually only a single sample is selected* and a single value of the statistic is observed, and we don’t know what the population distribution is.

Example 23.1. Recall that in the car dealership problem we were trying to estimate μ based on a random sample $X_1, \dots, X_n$ from a $\text{Poisson}(\mu)$ distribution. We considered a few different estimators of μ .

$$\begin{aligned}\bar{X} &= \frac{1}{n} \sum_{i=1}^n X_i \\ S^2 &= \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \\ \hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \\ \hat{\mu} &= \frac{n}{n+100} \bar{X} + \frac{100}{n+100} \quad (2.3)\end{aligned}$$

Let's see how these estimators perform when $\mu = 2$ for samples of size $n = 3$.

1. Describe in detail how you could use simulation to approximate the distribution of $\bar{X}$ for samples of size $n = 3$ when $\mu = 2$.
2. Conduct the simulation and approximate $E(\bar{X})$. Interpret this value.
3. Repeat parts 1 and 2 for S^2 .
4. Repeat parts 1 and 2 for $\hat{\sigma}^2$.
5. Repeat parts 1 and 2 for $\hat{\mu}$.
6. Identify if each estimator tends to overestimate μ , underestimate μ , or neither, when $\mu = 2$. But then explain why this information by itself is helpful, and what we should do next.

- $\hat{\theta}$ is an **unbiased** estimator of θ if

$$E(\hat{\theta}) = \theta \quad \text{for all potential values of } \theta$$

- Roughly speaking, an unbiased estimator does not tend to systematically overestimate or underestimate the parameter of interest, regardless of the true value of the parameter.
- The **bias (function)** of an estimator¹ is the difference between its expected value (over many hypothetical samples) and the true (but unknown) value of the parameter

$$\text{bias}(\hat{\theta}) = E(\hat{\theta} - \theta) = E(\hat{\theta}) - \theta, \quad \text{a function of } \theta$$

- If an estimator is unbiased its bias function is 0
- In general, the sample mean $\bar{X}$ is an *unbiased* estimator of the population mean μ .

$$E(\bar{X}) = \mu$$

- In general, the “ $1/(n-1)$ ” sample variance S^2 is an *unbiased* estimator of the population variance σ^2

$$E(S^2) = \sigma^2$$

Example 23.2. Consider estimating μ for a $\text{Poisson}(\mu)$ distribution. Determine which of the following estimators are unbiased estimators of μ based on a random sample from a $\text{Poisson}(\mu)$ distribution. If the estimator is not unbiased, identify and plot its bias function.

1. $\bar{X}$.

2. S^2

3. $\hat{\sigma}^2$. (Hint: $\hat{\sigma}^2 = \frac{n-1}{n} S^2$.)

¹We are only considering bias in *estimation*: is there bias in how we use the data to estimate the parameter? There could also be *bias in data collection*, in how the sample is selected and how the variables are measured. In this course we always assume that we have a “perfect” random sample from the population; in practice, that will never be true, but it might be a reasonable assumption based on how the data are collected. In any case, the bias function only quantifies bias in estimation, not bias in data collection.

4. $\hat{\mu} = \frac{n}{n+100}\bar{X} + \frac{100}{n+100}(2.3).$

Example 23.3. Continuing the $\text{Poisson}(\mu)$ problem. Let $\theta = e^{-\mu}$. We know $\bar{X}$ is an unbiased estimator of μ . We also know that $\hat{\theta} = e^{-\bar{X}}$ is the MLE of $\theta = e^{-\mu}$. Is $\hat{\theta}$ an unbiased estimator of θ ?

1. Describe in detail how you would use simulation to approximate the bias of $\hat{\theta}$ as an estimator of θ when $\mu = 2$.
2. Describe in detail how you would use simulation to approximate the bias *function* of $\hat{\theta}$.
3. Explain the reason why $e^{-\bar{X}}$ is a biased estimator of $e^{-\mu}$ even though $\bar{X}$ is an unbiased estimator of μ .
 - In general, if $\hat{\theta}$ is an unbiased estimator of θ , $g(\hat{\theta})$ is usually NOT an unbiased estimator of $g(\theta)$ (unless g is linear).
 - Because in general $E(g(\hat{\theta})) \neq g(E(\hat{\theta}))$.
 - For example, even though the “ $1/(n-1)$ ” sample variance S^2 is an unbiased estimator of the population variance σ^2 , the corresponding sample standard deviation S is a biased estimator of the population standard deviation σ

Example 23.4. Continuing the $\text{Poisson}(\mu)$ problem. Consider the estimator $\hat{\sigma}^2$. Recall that the bias function is $\text{bias}(\hat{\sigma}^2) = -\frac{\mu}{n}$. What happens to the bias function as n increases. What

does this mean?

- An estimator $\hat{\theta}_n$ is an **asymptotically unbiased** estimator of θ if $E(\hat{\theta}_n) \rightarrow \theta$ as $n \rightarrow \infty$ for all θ , that is, if $\text{bias}(\hat{\theta}_n) \rightarrow 0$ as $n \rightarrow \infty$ for all θ .
- For an asymptotically unbiased estimator, if the sample size is large $E(\hat{\theta}) \approx \theta$, regardless of the true value of θ . That is, if the sample size is large the estimator is “nearly unbiased”.
- MLEs are often biased, but in general, MLEs are asymptotically unbiased

23.1 Exercises

Exercise 23.1. Consider estimating the rate parameter λ of an Exponential population based on data $X_1, \dots, X_n$ from a random sample of size n . (For example, you want to estimate the rate at which earthquakes occur based on a sample of times between earthquakes.)

We have seen that the MLE of λ is $1/\bar{X}$.

1. Determine $E(\bar{X})$ in this case. Your answer should be an expression involving λ .
2. Is $1/\bar{X}$ an unbiased estimator of λ ? Justify your reasoning by considering expected values.
3. Explain in full detail how you could conduct a simulation to approximate the bias of $1/\bar{X}$ when $\lambda = 2$ for a given n .

4. Explain in full detail how you could conduct a simulation to approximate the bias *function* of $1/\bar{X}$ for a given n .

Exercise 23.2. Consider estimating the population mean μ and population variance σ^2 for a Normal population based on data $X_1, \dots, X_n$ from a random sample of size n . (For example, assume credit scores follow a Normal distribution and you want to use sample data to estimate the mean and variance of credit scores.)

We have seen the MLEs of μ and σ^2 in previous handouts.

1. Is the MLE of μ an unbiased estimator of μ ? If so, explain why. If not, find the bias function.

2. Is the MLE of σ^2 an unbiased estimator of σ^2 ? If so, explain why. If not, find the bias function.